

Expert Review of Medical Devices

ISSN: (Print) (Online) Journal homepage: https://www.tandfonline.com/loi/ierd20

Device profile of the MED-EL Cochlear Implant System for hearing loss: overview of its safety and efficacy

Uwe Baumann , Timo Stöver & Tobias Weißgerber

To cite this article: Uwe Baumann , Timo Stöver & Tobias Weißgerber (2020): Device profile of the MED-EL Cochlear Implant System for hearing loss: overview of its safety and efficacy, Expert Review of Medical Devices, DOI: 10.1080/17434440.2020.1781614

To link to this article: https://doi.org/10.1080/17434440.2020.1781614

Accepted author version posted online: 18 Jun 2020.

Submit your article to this journal

Article views: 1

View related articles

View Crossmark data

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ierd20

Check for updates

Publisher: Taylor & Francis & Informa UK Limited, trading as Taylor & Francis Group

Journal: *Expert Review of Medical Devices*

DOI: 10.1080/17434440.2020.1781614

Device Profile

Device profile of the MED-EL Cochlear Implant System for hearing loss: overview of its safety and efficacy

Uwe Baumann[1,*], Timo Stöver[2], Tobias Weißgerber[1]

1 Audiological Acoustics, ENT Department, University Hospital Frankfurt, Frankfurt am Main, Germany,

2 ENT Department, University Hospital Frankfurt, Frankfurt am Main, Germany

***Corresponding author:**

Uwe Baumann

uwe.baumann@kgu.de

University Hospital Frankfurt

ENT/Audiological Acoustics

Theodor-Stern-Kai 7

60590 Frankfurt M.

Germany

ACCEPTED MANUSCRIPT

Abstract

Introduction: Patients suffering from severe to profound hearing loss or even deafness can achieve a hearing improvement with a cochlear implant (CI) treatment that is significantly higher than the results achieved with conventional hearing aids. The CI system consists of an implantable stimulator, which is inserted retro-auricularly into the mastoid, and an externally worn processor unit, which provides the pickup of sound and processing of acoustic information as well as the power supply for the stimulator and internal current sources. The stimulator has an electrode array that is inserted into the cochlea.

Areas covered: This is a descriptive overview of MED-EL's multichannel CI system (MED-EL, Innsbruck, Austria), which was introduced to the European market in 1994. The continuing development of the implant as well as the external components is outlined and various other aspects (stimulation strategy, adaptation, results, reliability) are discussed.

Expert opinion: The strength of the company is the continuous pursuit of innovative ideas. This is evidenced by numerous innovations. The reliability of the implants has been continuously improved. The current SYNCHRONY models of the manufacturer show no indication of technically caused failures over an observation period of 5 years [1].

Keywords

Cochlear Implant; cumulative survival rate (CSR); profound hearing loss; deafness; hearing preservation, electric-acoustic stimulation (EAS), single sided deafness (SSD)

1. Introduction

The treatment of severe to profound hearing loss or even deafness has been revolutionized by the cochlear implant (CI) prosthesis. In contrast to other listening aids, CIs are used to bypass the harmed inner ear hair cells to enable the transmission of sound signals to the auditory nerve.

Introduced into clinical practice at the end of the 1980s, the awareness slowly developed that remarkably good speech discrimination can be achieved with direct electrical stimulation of the auditory nerve. In fact, most recipients of current devices use the telephone routinely. Nowadays, congenitally deaf children can be successfully supplied with CI systems in their first year of life if their hearing loss is recognized early by newborn hearing screening and there is no dysfunction of the auditory nerve.

With about 600,000 persons having received a CI in one or both ears, the number of recipients exceeds the number for any other neural prosthesis (e.g. deep brain stimulation, retinal or vestibular prosthesis, [2]). The restoration of function provided with a CI is superior to any other neural stimulation system [3].

The CI system consists of an internal and an external part, which can be connected to additional devices like remote controls, wireless microphones or audio streaming devices. The internal receiver is surgically implanted in the mastoid bone behind the ear, with electrodes inserted into the cochlea. The external processing device picks up sound via microphones. The sound is converted into an electrical signal which is sent to the electrodes in the cochlea. The electrodes then send electrical currents to the auditory nerve (Fig. 1).

In the past, patients with severe to profound hearing loss and residual hearing in the low frequency hearing were often not considered for implantation as the risk of losing this hearing was too high. Now, more and more patients are considering devices which use both electrical and acoustic

ACCEPTED MANUSCRIPT

stimulation thus providing high frequency information and preserving the low frequency hearing through a combination of hearing aid and cochlear implant technologies.

Implantation is also increasingly being considered for patients with unilateral deafness (single sided deafness, SSD). In some countries (e.g. Germany) the cost of providing the implant to patients suffering from SSD is reimbursed, because speech perception in background noise and acoustic localization are significantly improved.

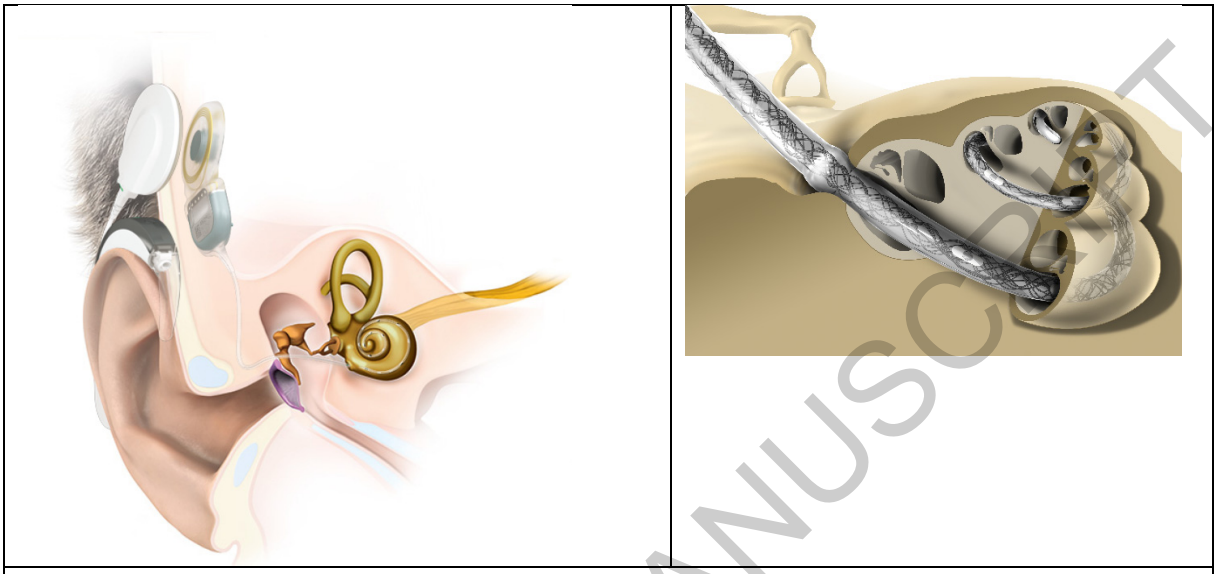


Fig. 1: SYNCHRONY 2 cochlear Implant System (MED-EL, Innsbruck, Austria). Left: External part (Sonnet 2 processor) with sending coil locked magnetically to the internal part consisting of implantable stimulator Mi1250 (titanium housing, receiving coil embedded in silicone, magnet) and electrode array (12 channels). Right: Cochlea sectional view with electrode array consisting of micro wires and platinum contacts embedded in silicone. Figures with kind permission by MED-EL.

ACCEPTED MANUSCRIPT

2. Body of review

This overview of the CI systems of the manufacturer MED-EL does not only refer to the current SYNCHRONY product line. The reason for this is that no other active implantable medical device is subject to comparable requirements with regard to the maximum required lifetime. Since nowadays congenitally deaf children are regularly treated with CIs already in their first year of life, a minimum implant operating time of significantly more than 20 years is desirable in order to minimize the number of revision surgeries required later.

Even the first product generation of the manufacturer's multichannel stimulators achieves a comparatively long lifespan. Young adults who have been fitted with one of these early models as a toddler must, like all other CI users, continue to undergo regular medical and technical examinations of their implant. For this reason, the following is an overview of the entire multichannel implant product line.

2.1. Development of the implant

The founders of MED-EL, Ingeborg and Erwin Hochmair, began developing cochlear implants at the Vienna University of Technology in 1975. Their aim was to enable users not only to perceive sounds but also to understand speech to a certain extent. The first implants had 2-4 electrode channels and analogue control. With the development of the digital high rate multichannel implant in the early 1990s, the corner stone for the later success of the company was laid.

2.1.1. Multichannel Implant

The first MED-EL high rate multichannel CI was introduced to the European market in 1994. The COMBI 40 is considered as being the first multichannel high rate CI, providing the CIS speech coding strategy on eight channels. Stimulation was accomplished with an intra-cochlear electrode (length 31 mm) and an external reference electrode, placed at the end of an extra lead under the temporalis muscle [4].

Place here fig. 2a-e
Fig. 2: Legacy implants. a) COMBI 40+ (1996), b) PULSAR (2004), c) SONATA (2006), d) Mi1000 (CONCERTO). Figures with kind permission by MED-EL.

To prevent direct current (DC) stimulation, capacitors were serially connected to the electrodes to block any unbalanced charge being delivered to electrodes. All later devices continued to use this technique. In addition, a check of the maximum charge is carried out to prevent overstimulation, whereby the charge density is kept below a certain level depending on the electrode material, size and shape.

Stimulation was realized with multiple current sources. A switching network connecting a power source with several electrodes, as used in the devices manufactured by COCHLEAR (Macquarie, Australia) at that time, was not necessary. Instead, multiple current sources are used sequentially or simultaneously, in which both the N-channel and the P-channel current sources are used to generate positive and negative phases of stimulation.

In the year 1996, the COMBI 40+ was brought into clinical application and extended the system to stimulate 12 channels. In addition, the size of the stimulator was reduced considerably (Figure 2.

In 2004, the PULSAR implant (PULSARci^{100}) was presented with advanced electronics and ceramic housing and extension of the backward telemetry (Auditory Nerve Response Telemetry, ART). The pulse stimulator capabilities were extended to generate *triphasic* or *precision triphasic* pulses, with later shown beneficial effects on the prevention of undesired co-stimulation of the facial nerve [5,6] as well as reduction of electrical artefact in ART-recordings [7].

With the introduction of the SONATA implant (SONATAti^{100}) in 2006, the housing material design changed from ceramics to titanium, presumably to improve impact resistance (Figure 2). In addition, the reference electrode was integrated as surface contact to the stimulator body.

The Mi1000 (CONCERTO) implant was part of the MAESTRO CI system, which was launched 2010 (Figure 2). The major visible design change was the reduction of the height of the titanium stimulator housing and the introduction of a bend protection at the outlet of the electrode lead.

The Mi1200 (SYNCHRONY) CI system was approved 2014 in Europe (FDA approval 2015). With the new design of the magnet, implant torque and the risk of demagnetization during MRI scans was greatly reduced. In addition, the magnet is surgically removable (special tool required) and can be replaced with a non-magnetic spacer intended to be used as placeholder for the regular implant magnet during MRI procedures, when in very rare cases a reduced image artifact is desirable.

The recently introduced Mi1250 (SYNCHRONY 2) implant includes a design change at the central exit of the electrode (Fig. 3).

ACCEPTED MANUSCRIPT

With the development of the CONCERTO titanium housing, all later implants are available also with PIN housing design. The PIN stimulator designs incorporates two titanium fixation pins on the bottom of the housing to prevent movement of the implant for long-term stability.

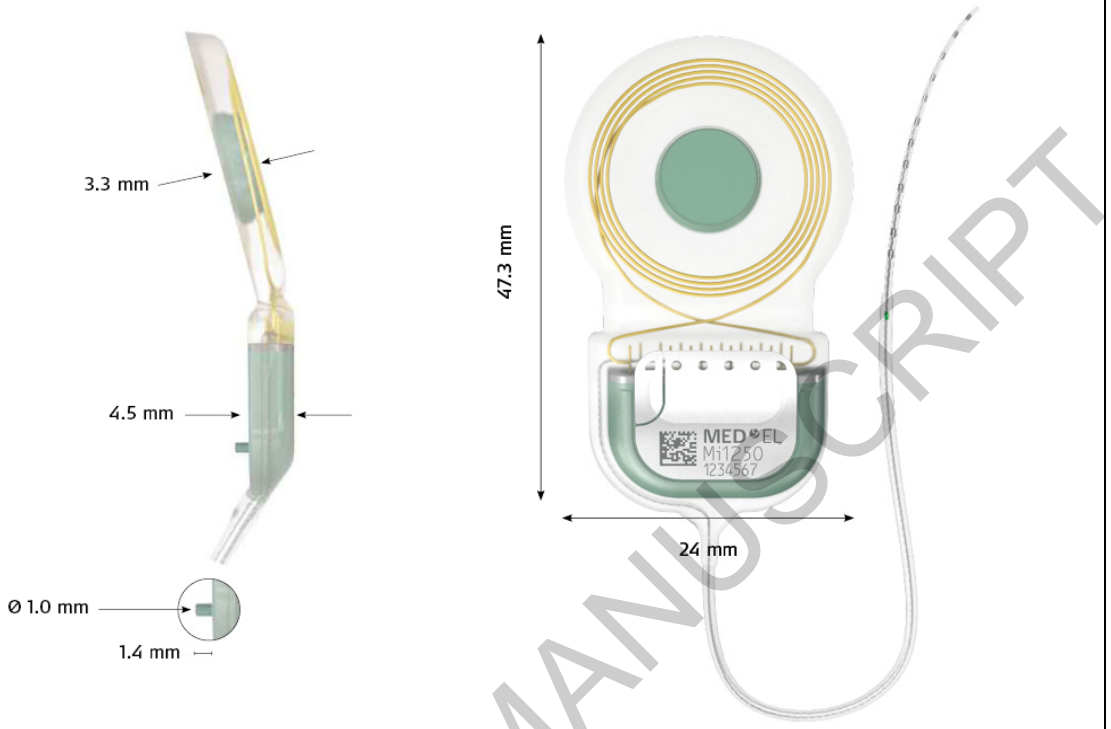


Fig. 3: Mi1250 (SYNCHRONY 2, MED-EL, Innsbruck, Austria) cochlear implant, size of stimulator 18.8 mm x 24 mm x 4.5 mm, coils 29 mm diameter x 3.3 mm thick. PIN variant with fixation pins for stability, weight 7.7 g, hermetically sealed titanium housing, impact resistance >= 2.5 Joule. Figures with kind permission by MED-EL.

ACCEPTED MANUSCRIPT

2.1.2. MRI safety

MRI compatibility is a safety issue that affects CI-patients lifelong and may lead to revision surgery in case of magnet dislocation following MRI [8]. Thus further technical improvements have been developed to reach MRI-compatibility up to the highest standard.

The Mi1200 and Mi1250 implants are MR conditional allowing MRI scans at 3.0T with the magnet in place. The magnet housing can also be substituted with a non-magnetic spacer for MRI head scans to reduce image distortion in the implant area (special surgical tool required). The removable magnet housing features a protective coating to prevent unwanted cellular adhesion.

In case of 1.0 T, 1.5 T and 3.0 T MRI systems, straight head orientation is required. The patient should not incline his/her head to the side; otherwise torque is exerted onto the implant magnet which could cause pain. In case of 0.2 T scanners, no specific head orientation is required [9].

For 0.2 T, 1.0 T and 1.5 T scans (see Table 1), only sequences in “Normal Operating Mode” with a maximum head Specific Absorption Rate (SAR) of 3.2 W/kg shall be used.

For 3.0 T scans the SAR limit must not exceed specific SAR values for specific anatomic regions to avoid any potentially dangerous heating at the electrode contacts. For the same reason head transmit coils or multichannel transmit coils must not be used in case of a 3.0 T MRI.

The manufacturer provides detailed information for radiologists and patients for all implantable devices on the internet [10].

2.1.3. Electrode portfolio

Compared to other manufacturers, the MED-EL electrode portfolio offers a very wide variety. On the one hand, the different lengths of the electrode carriers cover special anatomical situations, such as the very short designs used in cases of cochlear ossification or the electrodes of the FORM series, which are designed to seal the opening of the cochlea particularly effectively in the presence of the so-called "Gusher" syndrome (increased pressure of the lymphatic fluids of the inner ear). On the other hand, the different lengths of the electrode carriers are intended to provide tailor-made stimulation of the fibers of the auditory nerve in the cochlea, taking into account aspects of the individual size of the cochlea or the residual hearing left in the low-frequency range.

CLASSIC design

With the COMBI40+ implant, the CLASSIC electrode design was offered in three options: *standard* (31.0 mm/active length 26.4 mm), *medium* (24.0/20.9), and *compressed* (15.0/12.1). All CLASSIC electrodes are designed with 24 electrode platinum double window contacts (two electrodes connected in parallel), resulting in 12 electrode channels. Electrode numbering starts with electrode channel E1 located most apically and ends with E12 basally. As the active length of the electrode array varies between electrode configurations, the distance between electrode channels is adapted accordingly. The CLASSIC *compressed* design was originally introduced to support implantation in cases with cochlear malformation or intracochlear sclerosis, where drilling to open the basal part of the cochlea was necessary. The *medium* electrode design should improve the preservation of hearing in the apical part of the cochlea to achieve combined electrical and acoustic stimulation (adjunctive stimulation, see below).

FLEX design

Introduced together with the SONATA implant, the FLEX electrode design accomplished a thinner diameter especially at the tip to prevent structural damage of the delicate structures of the inner ear. The five most apical channels were designed as single electrode contacts, which allowed a very thin size of the tip diameter without loss of stability. FLEX arrays are currently available in five lengths (20 mm, 24 mm, 26 mm, 28 mm, 31.5 mm), which allows the adaption of electrode coverage to individual cochlear size and – if necessary – to the range of acoustic low frequency hearing (Fig. 4).

ACCEPTED MANUSCRIPT

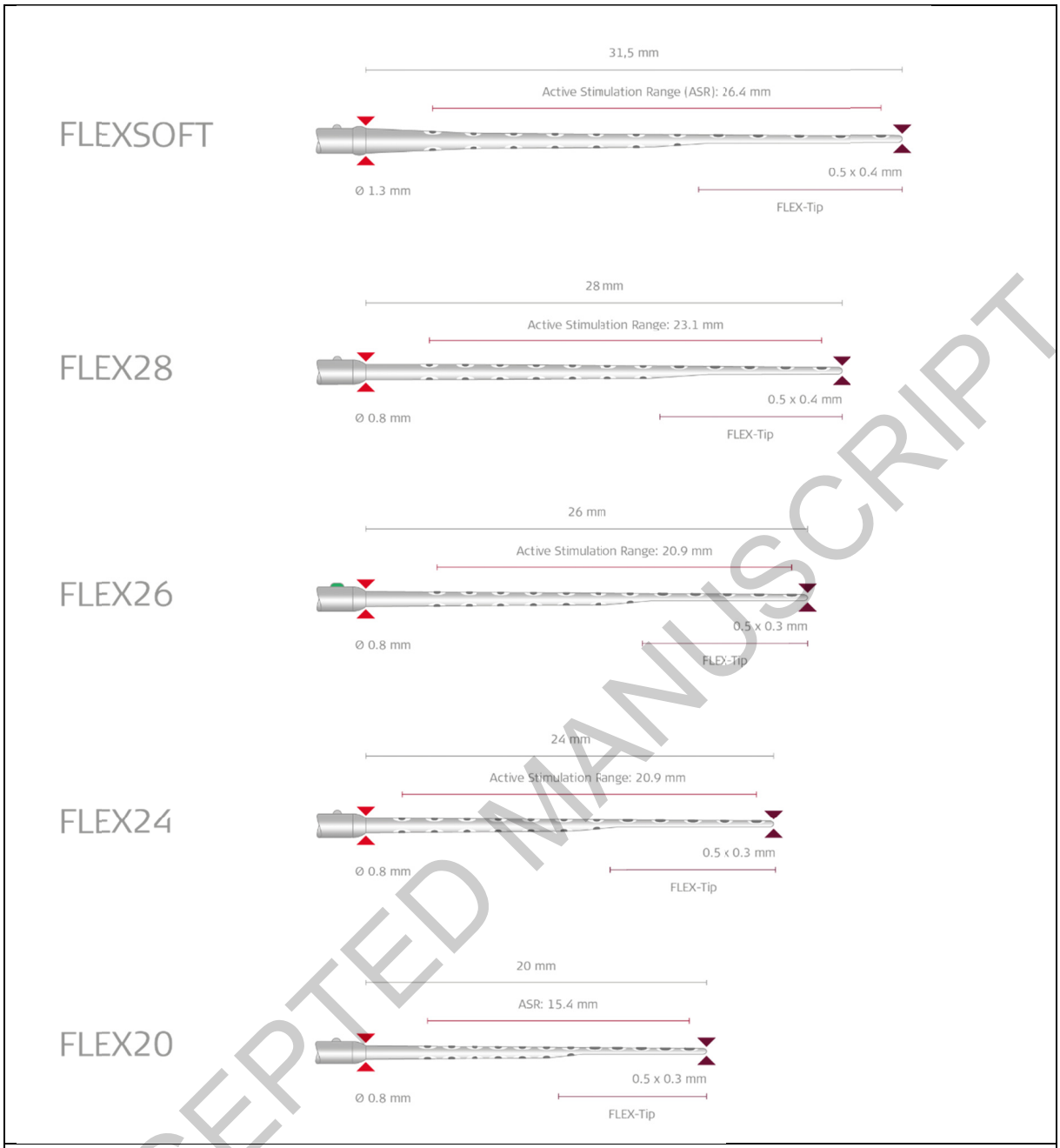


Fig. 4: FLEX electrode design options (MED-EL, Innsbruck, Austria). Platinum double contact windows up to electrode #7, five single electrode contacts at the tip. Figure with kind permission by MED-EL.

ACCEPTED MANUSCRIPT

Form design

FORM Electrode Arrays were introduced 2013 and designed specifically for malformed cochleae and for instances where leakage of cerebrospinal fluid is expected. Each FORM array features an integrated seal designed to close off the cochlear opening and / or making it easier for surgeons to apply additional tissue for sealing the area once the electrode array has been inserted.

Insertion test device

Preoperative imaging cannot always be used to reliably determine the actual accessibility of the cochlear lumen. In difficult cases as inner ear malformations or vestibular schwannoma, the use of intraoperative intra-cochlear test electrodes is recommended. The insertion test tools may be used during inner ear surgery to assess the condition of the cochlea. This information can be used for

selecting the best suited cochlear implant electrode variant in terms of insertion depth vs. number of contacts inserted [11].

More information about the MED-EL electrode portfolio is available in [12].

2.2. Development of processing devices

Due to the continuing development of device technology, especially the miniaturization of the components, the processor units used to control the implanted stimulator could be reduced in their external dimensions. At the same time, the energy consumption of the system was reduced, allowing the development of devices worn behind the ear. Compared to conventional hearing aids, however, the energy requirement is comparatively high, as there is a considerable loss due to the transcutaneous wireless energy transmission via radio frequency. For example, the SONNET processor with two 675 zinc-air batteries provides a runtime of between two and three days in practice, while conventional hearing aids have a runtime of up to 14 days depending on the type of device.

2.2.1. Body-worn and BTE devices

With the introduction of the first MED-EL high rate multichannel CI, body worn processing devices were the only way to power the implanted stimulator and perform complex signal processing. With the design of dedicated application specific integrated circuits, power consumption was reduced to allow the design of the first multichannel Behind-the-Ear (BTE) processor in 1999 (TEMPO+). Driven by three 675 zinc-air batteries, on average 30 hour usage was accomplished. Further downsizing of the electronics allowed a smaller BTE design (OPUS 2), which was launched in 2006. Since the OPUS 2 processor had no knobs for sensitivity or program adjustment, a remote control (FINETUNER) was introduced for this purpose. The outer dimensions of the OPUS 2 were reduced further by the development of a smaller battery compartment comprising only two 675 zinc-air batteries.

Together with the introduction of the Mi1200 implant, a new processing device with directional microphone design was introduced in 2014. The Me1310 (SONNET) processor extended protection against humidity and dust. Depending on surrounding noise level, the automatic engagement of directional microphone sensitivity was integrated as well as automatic wind noise reduction into the signal preprocessing. One year later, rechargeable batteries and a new design of the transmitting coil were introduced. Wearing options comprising clips, special battery compartments and extended cables for babies and small children were deliverable since 2017. In addition, battery compartment integrated wireless receivers were available to connect Roger 21 wireless microphones (Phonak, Stäfa) or other accessories. The Me1310 device enabled data logging to monitor average wearing time as well as program settings. With introduction of the DL transmitting coil, the monitoring of the contact and the integrity of the digital connection to the stimulator was made available.

The latest Me1510 (SONNET 2, 2019) audio processor offers noise reduction, adaptive signal processing (Automatic Sound Management, ASM) and allows setting of volume, sensitivity and program with smart phone devices. Data logging was extended to include information about input sound levels (dB) and accessory usage. Together with the launch of Me1510, a new remote control (FINETUNER ECHO) combining an electric ink display with an integrated processor test ability was introduced (Fig. 5 left).

ACCEPTED MANUSCRIPT



Fig. 5: Left. Me1510 processor (Sonnet 2, width 37.4 mm, thickness 5.9 mm) with DL-coil, Audiolink, AudioKey App and FineTuner Echo. Right: Me1150 (Rondo 2, 46.8 mm x 35.8 mm, thickness 12.1 mm) single unit processor with charging device. Figures with kind permission by MED-EL.

A dedicated wireless streaming device (AUDIOLINK) is available since 2019, which is compatible with SONNET and SONNET 2 processors. The device provides smart phone streaming via Bluetooth, offers a direct audio input (e.g. connection of TV) as well as a built-in microphone intended to use as a remote microphone.

2.2.2. Single-Unit devices

With the introduction of the RONDO in 2013, power consumption and weight of the processor was reduced to form a single-unit processor which is worn directly over the implant stimulator. Results obtained with this device were comparable to the BTE device [13]. This option is usually chosen for aesthetic reasons, but is also recommended for patients wearing glasses and/or with small ears to avoid pressure points or discomfort.

In 2017, the Me1150 (RONDO 2) replaced the RONDO and integrated a lithium battery and wireless charging (Fig.4 right). Usability was further improved by reduced weight and size. However, directional microphones as implemented in the SONNET device, were not introduced.

2.2.3. Adjunctive stimulation

In case of useful residual low frequency residual hearing, combined electric and acoustic stimulation (EAS, or hybrid stimulation) in the implanted ear is beneficial for this group of patients. Combined EAS can improve speech reception substantially as well as music reception and sound localization abilities [14].

The first CI processor combining electric and acoustic stimulation (DUET) was introduced in 2005. Until then, an ITE (in the ear) hearing aid had to be used in addition to the CI processor, which had to be fitted separately to the residual hearing. In 2009, the second generation EAS processor DUET 2 was released. With the introduction of the Me1320 (SONNET EAS) processor, an EAS supporting device version was made available (Fig. 6). FDA approval of the EAS method and Me1320 devices was granted in 2016 [15].

Since the acoustic driver is integrated into the housing of the Me1320 processor, custom earmolds are required to transmit the acoustic signal to the eardrum. Acoustic stimulation range is up to 2 kHz and the acoustic amplification is provided with a maximum gain of 48 dB. Maximum power output (MPO) is limited to 118 dB SPL. Compression is set in 7 low frequency bandpass channels [16].

ACCEPTED MANUSCRIPT

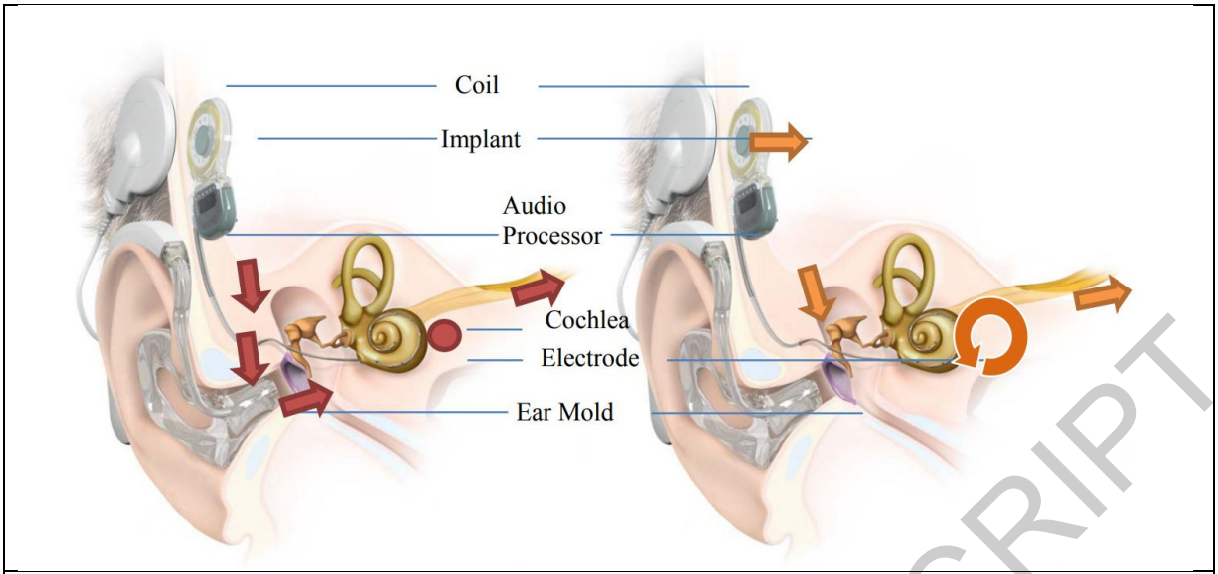


Fig. 6: Combined (adjunctive) stimulation, principle of operation. Left: acoustic stimulation, lower frequencies. Right: electric stimulation to cover the higher frequencies. Me1320 processor, Mi1200 implant, FLEX24 electrode. Figures with kind permission by MED-EL.

2.2.4. Water activities

Although resistance against humidity improved with the introduction of the SONNET device (degree of protection IP54 according to ISO 20653:2013-02), for swimming or other water activities an additional waterproof cover *WaterWear* is recommended (IP68, available for RONDO 2, RONDO, SONNET 2, SONNET, OPUS 2). *WaterWear* is fixed in place with an adhesive strip. *WaterWear* covers are reusable and can be used up to three times. The usage of zinc-aire batteries is not possible, either rechargeable, alkaline or silver-oxid batteries are required [17].

2.2.5. Backward compatibility

Although the shape of audio processors and transmitting coils changed considerably in the past 25 years since the introduction of the first MED-EL multichannel cochlear implant, all external devices are compatible to all implants (note: not compatible to active middle ear implants or bone conduction devices, BONEBRIDGE, of the same manufacturer).

2.3. Back telemetry

As early as with introduction of the COMBI 40 implant, MED-EL incorporated the function of back telemetry to check the status of the internal unit. This function of back telemetry is critical to ensure that the internal circuit works in the proper state and can correctly execute commands sent from the external unit. The other function of back telemetry is to measure and monitor information regarding the electrode-tissue interface, including electrode impedance and electrode field potential. Later, with the introduction of the PULSAR stimulator, the measurement of neural responses (compound action potentials, CAP) was also introduced.

The various applications of the back telemetry demand very different requirements on the measuring amplifier integrated in the implant, as a large input voltage range must be detected (microvolts to volts). A programmable amplifier is typically used, which determines the input sensitivity in the corresponding area of the AD converter. To transmit the information obtained in the implant to the external coil, modulation of the load on the receiver is applied, similar to the method used in RFID applications, whereby the load on the internal coil is changed in such a way that small changes in the amplitude of the RF signal can be detected at the external coil. For load modulation, the quality factor of the RF-receiver circuit within the implant is reduced, and this reduction is detected outside.

ACCEPTED MANUSCRIPT

In the application, digital data are transmitted by means of load modulation at a rate of 300 kbits/sec. The duration of the read-back of the 2048x1-RAM at 300 kbit/sec is about 7 ms. Thus, together with the duration of 1.7 ms for the measurement window, the maximum repetition rate for EAP-measurements is more than about 100 Hz [18].

The advantage of the RFID method is that only one pair of coils is required for forward and backward transmission. Disadvantages are the high complexity of the RF decoder in the external coil and the fact that simultaneous transmission over the outward and return path is not possible [19].

Electrically evoked CAP (eCAP) measurements allow longitudinal monitoring of the healthiness of the auditory nerve fibers and can also be used to determine the initial stimulation current levels that are required to program the CI processor. Generally, eCAP measurements are affected by noise and artifacts, therefore signal processing strategies such as averaging over repeated stimulation parameters are necessary. In order to increase the accuracy of the eCAP threshold determination, as well as to allow automatic detection of thresholds, a procedure named “AutoART” was introduced with the MAESTRO 7 clinical fitting software. Thereby, instead of averaging over repeated recordings with the same stimulation intensity, moving averages are computed over a small range of approximately identical stimulation levels. To exclude recordings that do not contain any potential nerve responses, AutoART uses an automatic classifier that estimates the firing probability from a given eCAP recording, and evaluates whether that probability lies within human physiological range. Determination of the eCAP threshold based on a sigmoidal fitting of the amplitude growth function allowed a robust and precise determination being indistinguishable from evaluations by human experts [20].

2.4. Stimulation strategies

The strategy for the transmission of signal envelope information (continuous-interleaved-sampling, CIS, [21]) was first implemented with the introduction of the COMBI40 implant and increased from eight to twelve channels with the COMBI40+ device. Initially the overall bandwidth was 350 Hz-5,500 Hz, later processors extended the range to 70 Hz-8,500 Hz. Each processing channel includes a band-pass-filter, followed by an envelope detector (implemented first as rectifier and later more complex as the output of a Hilbert transform). After extraction of the temporal envelope, the signal is compressed using a logarithmic or power law function in order to map the large acoustic dynamic range to the comparatively small range of electrical stimulation (5-20 dB for pulses vs. 90 dB or more for normal hearing). The compressed envelope signal modulates a carrier of balanced biphasic pulses with a carrier rate between a few hundred and a few 1000 pulses per seconds (pps, clinical range: 700 – 1500 pps). The output of each channel is directed to the intra-cochlear electrodes with lower frequencies send to apical and higher frequencies to basal electrodes. In order to avoid the simultaneous superimposition of the electrical fields emitted by the stimulation electrodes, the pulse patterns are transmitted sequentially. Thereby, the CIS strategy prevents simultaneous channel interactions and at the same time transmits the temporal envelope for each band, given that the rate of the carrier is sufficiently high. Typically, the cut-off frequency of the low-pass filter for obtaining the temporal envelope is set to 400 Hz or below. Due to the sampling theorem a minimum carrier rate of 800 Hz is required for a sufficiently precise sampling of the temporal envelope.

As an extension to CIS, MED-EL introduced the channel-specific sampling sequence (CSSS) developed by Clemens Zierhofer under the name Fine Structure Processing (FSP) [22]. A CSSS pulse train is generated with every positive zero crossing of the bandpass filter output signal. The length of the CSSS pulse sequence is determined by the upper cut-off frequency of the bandpass filter in order to avoid overlaps during successive zero crossings. As a result, the repetition rate of the CSSS pulse pattern of one or more apical electrode channels is given as a function of the frequency of the

ACCEPTED MANUSCRIPT

associated bandpass filter output signal profile and thus of its instantaneous frequency. Instead of stimulation pulses with a normal pulse width (approx. 20 µsec), stimulation packets with shorter pulses are used in order to achieve sufficient loudness (Fig. 7).

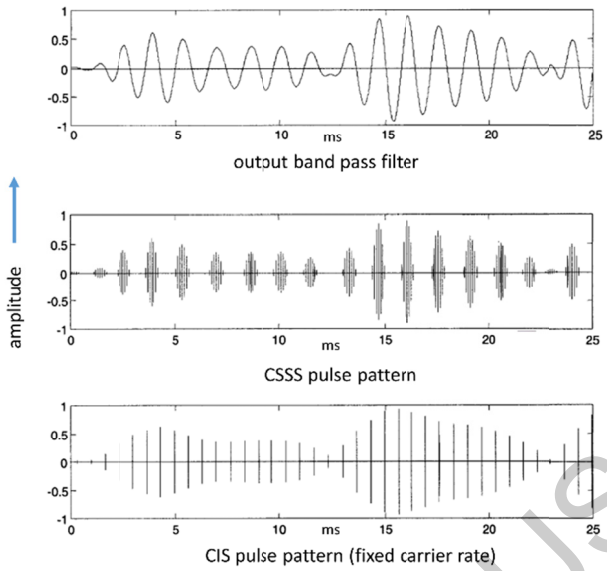


Fig. 7: Fine Structure Processing strategy. Top: output band pass filter. Middle: electric pulse pattern consisting of 6 kHz biphasic pulse packets (envelope Gaussian shaped), each packet in correspondence to zero crossing of the band pass signal. Bottom: Continuous interleaved stimulation (CIS), biphasic pulses, sampling of envelope information. Adopted from [26].

The number of FSP channels is limited by the required stimulation current; approximately 1-3 electrodes can be controlled with the FSP strategy. All other electrode channels are stimulated using the CIS method described above. With the FSP strategy, the range of low frequency transmission was extended from 250 Hz (CIS) down to 70-100 Hz in order to be able to transmit fundamental frequency information of male speakers. Improved transmission of music and pitch information was expected due to incorporation of rate information using the CSSS method. This was demonstrated in a majority of the subjects in a comparative study [23]. The results of a long-term follow-up observation presented by Kleine Punte and colleagues showed a significant improvement in speech reception in noise after 12 and 24 months of listening experience compared to a control group supplied with the (HD)-CIS strategy [24].

The FSP strategy was later supplemented by the FS4 procedure (introduced with the MAESTRO System Software 4.0, 2010). A maximum of 4 apical channels (upper limit frequency bandpass 950 Hz) are controlled with the CSSS method, in which a higher sampling rate (6,000 pps) of the fine structure channels was realized in order to ensure precise sampling of the filter output signal. If two medial or basal electrode channels are deactivated (e.g. electrodes not inserted into the cochlea), the number of CSSS channels is limited to three in order to counteract the resulting broadening of the remaining 10 bandpass filters. If events occur in parallel during the zero crossing detection, the channel with the greater amplitude is selected [25].

With the FS4-p strategy, a very similar procedure was presented, which additionally controls the distribution of the electric field in the event of a temporary coincidence of adjacent zero crossings through channel steering. The strong summations that may arise here are to be reduced by means of compensation currents, which are calculated using a model assuming an exponential field distribution [26,27].

ACCEPTED MANUSCRIPT

2.5. Clinical profile

The current MED-EL Flex-Electrode Portfolio provides a wide range of electrode lengths between 20 and 31.5 millimeters. This allows to tailor the electrode choice to the individual anatomy of the patient. Thus pre-surgical planning today includes the assessment of the anatomical dimensions of the cochlea length in order to personalize the electrode length. This is of special importance to combine hearing preservation with adequate electrical stimulation.

2.5.1. Surgical aspects

The procedure for the implantation of the stimulator is described in the surgical guideline [9]. Summarized, a curved incision is made in the retroauricular region, and a skin flap is prepared. After drilling the mastoidectomy followed by a posterior tympanotomy, a periosteal pocket for the implant is prepared using a periosteal elevator. An implant template can be used to mark the flatness on the skull and the correct position for the implant bed. For protection and placement of the electrode lead, a smooth channel has to be drilled in the bone leading to the mastoid. Access to the scala tympani of the cochlea is regularly established either by a cochleostomy near the round window or by opening the round window itself after preparing a clear view of the round window. Either a micro-lancette or a micro-hook can be used to open the cochlea. Then, insertion of the electrode lead should approach the anterior portion of the basal turn at an angle so that it slides along the lateral wall of the scala tympani. This procedure, known as tangential insertion, facilitates deep electrode insertion. During insertion it is essential that the electrode contacts are not mechanically damaged and that no excessive force is used. The most commonly used technique for the fixation of the stimulator is to use a tight periosteum pocket.

With the introduction of the SONATA implant, the housing material of the stimulator was changed from ceramic to titanium. The safety of the new design was examined in a study on 30 adults and 20 children [28]. The device has proven to be effective in that it is stable in the implant bed and a minimally invasive procedure can be performed. The device proved also to be safe after medium-term use and no unexpected adverse events were reported.

As more and more surgeons decided to abandon sutures or other measures like drilling a flat stimulator bed for fixation of the stimulator housing, the manufacturer introduced with the CONCERTO PIN device two 1.4 mm long titanium pins on the bottom of the housing to facilitate the immobilization of the implant to the skull. Fixation of the stimulator by drilling holes in the skull to receive the pins should be done in such a way that no postoperative movements occur. Continuous movement may result in mechanical fatigue and subsequent premature failure of electrical connections. Results of a study concerned with the safety and stability of the CONCERTO PIN (n = 100 recruited patients) demonstrated stable implant fixation and no device related adverse events over a follow-up period of 6 months [29].

Long-term electrode dislocation is a potential risk of straight electrode designs, in comparison to pre-curved electrodes [30]. Several techniques have been described to reduce the risk of electrode dislocation successfully as the use of a fixation clip or the adaptation of the surgical procedure. A new surgical technique has recently been described to minimize the risk of electrode dislocation. The fixation in a bone groove in the chorda-facial angle was proven as an effective measure against electrode migration. Proper surgical execution is mandatory to ensure a tight and enduring electrode fit [31].

Regarding the mechanical design of the electrode lead, a retrospective study was conducted on 1,320 patients (1,722 ears) implanted with CI stimulators from various manufacturers. No incidents of electrode misalignment or electrode tip roll- or fold over have been reported in 883 MED-EL electrodes [32].

ACCEPTED MANUSCRIPT

2.5.2. Hearing and structure preservation

Hearing and structure preservation is of great importance in patients with residual low frequency hearing. Adjunctive stimulation (electric acoustic stimulation, EAS) combines electric and acoustic stimulation in the same ear and is beneficial for speech reception as well as music reception and localization abilities [33–35]. Helbig and coworkers propose the audiometric indication for hearing preservation with subsequent EAS up to 65 dB HL at 500 Hz [36].

In order to define "hearing preservation", a classification system was proposed for all cochlear implant users with measurable pre-operative residual hearing. The universal reporting standard was designed to compare the results of studies and meta-analyze their data [37].

The basic design of the Flex-Electrode portfolio provides a high probability of atraumatic insertion resulting in structure preservation and finally hearing preservation. This presents the basis for electric acoustic stimulation.

In addition to individual factors such as the etiology of the hearing disorder, the surgical technique or the experience of the surgeon, the design of the electrode array significantly influences postoperative hearing ability [38]. Due to the decreasing diameter of the cochlea towards the apical end and the necessary insertion force increasing with increasing insertion angle, there is a correlation between postoperative hearing loss and electrode length. Therefore, the electrode length and, if necessary, the insertion depth should be determined based on the analysis of the preoperative high-resolution X-ray diagnosis of the individual cochlea. A new tablet-based software package is available for this purpose, which calculates angular insertion depth (AID) and cochlear duct length (CDL) based on postoperative computed tomography. The AID and CDL are used to estimate the cochlear place frequency for each electrode on the basis of the Greenwood function. According to the results of this procedure, the most suitable electrode length can be chosen to allow hearing preservation in the lower frequency range [39].

Combining FLEX-electrode design, round window membrane surgical approach to the inner ear, perioperative application of corticosteroids and antibiotics, and adaptation of the surgical technique (e.g. slow electrode insertion) leads to convincing hearing preservation results.

With regard to the risk of deafness due to the insertion of the electrode into the cochlea, a rate of currently about 10 percent can be assumed [15]. A reduction of this rate is targeted by the development of methods to monitor the function of low frequency hearing during electrode insertion [40]. Besides telemetric measurements of the distribution of the electric field in the cochlea, electrophysiological recordings are used for this purpose [41].

Long-term hearing preservation is feasible in a subset of patients. No correlation of total hearing loss with etiology, electrode design, or surgical approach was evident so far. Apart from individual effects of structural damage or inflammation, genetic factors are suggested to influence hearing preservation. Cases with total hearing loss still demonstrated successful speech perception in long-term monosyllable recognition scores [42].

2.5.3. Fitting

Because each patient is different in terms of e.g. etiology and sensitivity of the auditory nerve, the parameters of electrical stimulation need to be customized to meet his or her conditions and needs. With MED-EL devices, the MAESTRO fitting program (current Version 8.0.3) is used to interface between a PC and the external CI-processor. In addition, a dedicated programming interface (MAX, connected via USB port) is needed to provide two-way communication during fitting sessions, which is generally conducted one month after surgery or as early as two days after surgery (early fit [43]). The fitting program collects patient-specific information, e.g. processing strategy (CIS, FSP, FS4 etc.)

ACCEPTED MANUSCRIPT

and the setting of threshold (THR) and most comfortable loudness (MCL) values on all electrodes. Once these stimulation parameters are set, the fitting program stores the information as a so-called “map” on the PC, which is then downloaded to the CI-processor memory.

Figure 8 shows a section of the MAESTRO fitting program. Electrodes are numbered from 1 (most apical, lower frequencies) to 12 (most basal, higher frequencies). The electrodes can be flagged (here electrode 12 as example), if they are malfunctioning electrically (e.g., short or open circuitry) or cause unpleasant or unwanted problems such as pain or facial nerve stimulation. To create the map on all other useful electrodes, both the lowest current level inducing threshold (THR) and the highest level inducing most comfortable loudness (MCL, typically loud but not too loud perception of test stimulus) are measured and recorded. The bars displayed in the fitting window indicate graphically the setting of THR and MCL parameters. The unit [qu] is reflecting the amount of electrical charge. Charge units are defined as the product of the amplitude of the stimulus current and the pulse width divided by 1000 (charge unit = (amplitude of current x pulse width)/1000). The system automatically adjusts the amplitude of the stimulation current and/or the pulse width to achieve the desired perception of loudness, with the goal of avoiding voltage compliance limits and maximizing the stimulation rate.

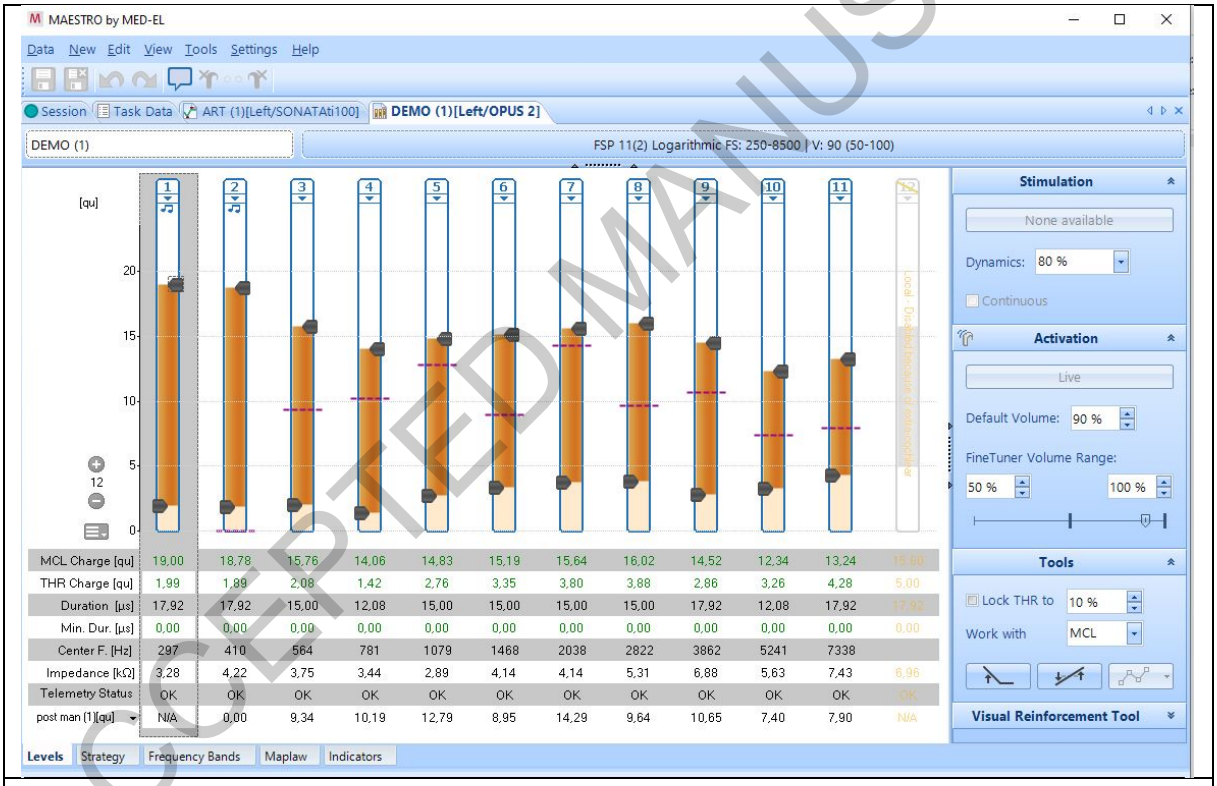


Fig. 8: Clinical fitting software application MAESTRO 8.03, window section with options for setting specific parameters of electrical stimulation. See text for further details.

The electrical parameters usually change with different processing strategies and electrode configurations, and sometimes may change with time and experience. Therefore, measurement and monitoring of these parameters periodically is mandatory. Four different maps can be downloaded as separate programs to current CI-processors. The fitting process can be time consuming, lasting from 20 minutes to 2 hours, depending on the device complexity, patient condition, and audiologists’ skill levels.

2.6. Hearing rehabilitation results

The success of hearing rehabilitation with a CI system depends on a large number of influential factors. The integrity of the auditory nerve as well as the number of effectively usable electrode

ACCEPTED MANUSCRIPT

channels can be regarded as the most important prerequisites [44,45]. Furthermore, the onset of hearing loss and the duration of deafness are of major importance in the CI rehabilitation of adults [46]. If complete deafness has existed since birth in adults and no oral communication is developed, an outcome limited to the detection of sounds can be expected. In congenitally deaf or very early deafened children in the absence of further disabilities, a very satisfying result can be expected with immediate implantation of a CI and subsequent specific rehabilitation of hearing and spoken language.

2.6.1. Adults (post lingual deafness)

First results on safety and effectiveness of the early COMBI 40 implant were reported from a cohort of post lingual deafened adults [47]. The results showed that complete insertion of a comparatively long electrode (maximum insertion depth 30.8 mm) was achieved in 42 of 60 cases. In only 5 cases, one or two electrode contacts remained outside the cochlea. A very low rate of medical/surgical complication was reported. At the time of the report, 25 subjects had completed their 1-year evaluation. Mean sentence recognition increased up by then to 97 % and the monosyllabic word understanding was 54 % and more than 68 % for the top quarter group. Safety and efficacy of the COMBI 40 system in adults was clearly demonstrated. Early after switch-on, a very rapid learning curve with the implant was observed in most patients.

Outcomes in noisy environments

The current Me1310 / Me1510 (SONNET / SONNET 2) processor is equipped with a dual-microphone system that allows three different directional microphone characteristics (*omnidirectional*, *natural* and *adaptive*). The effectiveness of the implemented fixed and adaptive beamforming in an adult user group has been shown [48,49] In addition, wind noise reduction is also integrated in three settings: *off*, *mild* and *strong*. The outcome of these measures for improvement of speech perception in complex acoustic environments was studied recently by Hagen et al. [50]. Compared to the predecessor OPUS 2 with only omnidirectional microphone sensitivity, the directional microphone setting (natural or adaptive) allowed clinically significant better speech perception in test conditions with the speech signal arriving from frontal direction and the noise signal from three loudspeakers located at +90,+180, and -90 degrees. Compared to an omnidirectional microphone, directional microphones provided a 3.2–4.7 dB benefit. Regarding speech understanding in wind, directional microphones are more sensitive for disturbances. Wind noise reduction in the SONNET can, depending on the setting, mitigate the negative effects of using directional microphones.

Outcomes adjunctive stimulation

A prospective study on safety and efficiency of the Mi1200 FLEX24 implant and Me1320 (SONNET EAS) processor was carried out in 14 clinical centers located in the U.S.A. [15]. The data supports that the MED-EL EAS system provides significant benefit compared to the preoperative acoustic-only condition, and that electric stimulation alone also provides benefit over the preoperative acoustic-only condition (mean improvement CUNY sentences: 18.4 % after 12 months). 85 % of the subjects exhibited better performance on speech recognition at the 12-month interval compared to the preoperative baseline (e.g., CUNY post EAS vs. pre acoustic-only). Therefore, the MED-EL EAS system is expected to improve speech recognition in terms of CUNY sentences and CNC words for a majority of the indicated population.

In terms of the efficacy of the directional microphones integrated in the Me1320 processor, Weissgerber and coworkers reported no significant difference between DUET 2 and Me1320 processor using the omnidirectional microphone in quiet and in noise. There was a significant improvement in SRT with settings *natural* (2.2 dB) and *adaptive* (3.6 dB). No detrimental effect of the WNR algorithm on speech perception was found in the absence of wind [51].

ACCEPTED MANUSCRIPT

Outcomes music perception

Despite advances in technology, the ability to perceive music remains limited for many cochlear implant users. Technological, biological, and acoustical constraints make music an especially challenging stimulus. The limitations of cochlear implant devices, which have been optimized for speech comprehension, become evident when applied to music, particularly with regards to inadequate spectral, fine-temporal, and dynamic range representation [52].

E.g., psychoacoustic pitch matching experiments in cohorts of MED-EL CI-users suffering from single sided deafness (SSD) examined electrically induced place-pitch or rate-pitch perception [53-55]. The task of the subjects was to compare the pitch generated by the electric stimulus with an acoustic stimulus in the normal hearing ear. The results showed, that the majority of subjects did not estimate the electric pitch in a manner as normally expected, since large deviations from the normal tonotopic pitch perception were frequently observed.

A study on music perception in a cohort of users of adjunct acoustic stimulation compared EAS users with two control groups: unilateral CI users and normal-hearing (NH) listeners. Groups were matched according to age and musical experience before hearing loss. No difference was found on rhythm perception, chord discrimination, dissonance rating, and emotion rating subtest performance between groups. Mean frequency discrimination scores were significantly better in EAS participants than in CI participants and not significantly worse than in NH participants. However, the EAS and CI groups scored similarly (significantly worse than NH participants) on both instrument detection and identification. The authors concluded that acoustic hearing in the low frequencies is helpful for music perception, though not the only important factor [34].

2.6.2. Children (prelingual deafness)

Of the many variables that contribute to an individual child's development after implantation, age at implantation, the presence of medical comorbidities, social determinants of health, and the provision of bilateral versus unilateral hearing are those that can vary widely and have consistently shown clear impacts. Specifically, age of implantation is crucial to reduce effects of deafness on the developing auditory system and capture the remarkable plasticity of early development. Language development after cochlear implantation requires therapy emphasizing hearing and oral communication, education, and other support which can be influenced by known social determinants of health; specifically, outcomes in children decline with reductions in socioeconomic status and levels of parental education. Medical co-morbidities also slow rates of progress after cochlear implantation. On the other hand, benefits of implantation increase in children who are provided with access to hearing from both ears. In sum, cochlear implants promote development of hearing in children and the best outcomes are achieved by providing early access to sound in both ears. These benefits can be limited by known social determinants of health which restrict access to needed support and medical comorbidities which add further complexity in care and outcome [56].

If the conditions are good, an almost normal development of oral speech can be expected in children with early, preferably simultaneous bilateral CI provision. As an example, the world-wide first early bilateral implanted child (1st ear implanted age 24 months in year 1996, 2nd ear implanted age 48 months) was able to attend regular schools and is now doing his Master in Computational Science (interview in [57]).

2.7. Reliability

Data on the reliability of later MED-EL implants (Sonata, Concerto and Synchrony) is available from the manufacturer and listed in table 1 [1]. Excluding accidents or medical reasons for removal of the implant, less than 1 % of all implanted devices have shown failures, even after up to 13 years of use. Including accidents and medical problems, the failure rate is between 3 to 4 %.

ACCEPTED MANUSCRIPT

Period	5 years			longest observation period		
Device	DR	DR+ACC	DR+ ACC+MED	DR	DR+ACC	DR+ ACC+MED
Sonata	99.95%	98.48%	97.28%	99.85%	98.14%	96.27%
Concerto	99.94%	98.57%	97.09%	99.90%	98.43%	96.48%
Synchrony	100.00%	98.76%	96.91%	x	x	x

Table 1: Cumulated survival rate (CSR, percentage) for MED-EL Sonata, Concerto and Synchrony implants as of October 2019 according to manufacturer's data [1]. DR: device related failure, ACC: accident related failure, MED: medical reason for explant. Longest observation period: Sonata 13 years, Concerto 9 years, Synchrony 5 years.

The reliability of earlier MED-EL devices implanted in between 2003 and 2013 is subject of a recent multicentric study of the HEARRING group [58]. The data presented there included 5462 children and 6200 adults collected across a ten year period. The authors presented figures on cumulative failure percentage (complement to cumulative survival rate, CSR). In order to compare these results with the data provided by the manufacturer for the newer implants, data on CSR was extracted and listed in table 2. Data is divided into failures occurring in adults (>= 10 years of age) and children (< 10 years of age). The oldest generation of cochlear implants (C40+ and PULSAR) had the highest failure rates and the largest difference in terms of CSR compared to adults. This clearly reflects the improvement in the technology over time; such as changes in housing made by the company manufacturer, addressing device vulnerabilities or changes in cochlear implant production. The titanium housing of the newer generation implants SONATA and CONCERTO seem to be more robust than the ceramic housing of the C40+ and PULSAR.

Overall failure rate for all subjects and devices was 2.41 % with a mean annual failure rate of 0.63 %. Mean annual failure rate is within the range reported by Wang et al. [59].

Period	5 years		LOP	
Device	DR	DR+ ACC+MED	DR	DR+ ACC+MED
Children				
C40+	96,5%	92.8%	93.4%	88.2%
Pulsar	96.8%	94.8%	93.2%	87.5%
Sonata	97.8%	96.9%	97.8%	96.9%
Concerto	99.2%	99.0%	x	x
Adults				
C40+	99.7%	97.9%	99.2%	96.5%
Pulsar	99.8%	98.6%	99.7%	98.2%
Sonata	99.9%	99.0%	99.9%	99.0%
Concerto	99.8%	99.0%	x	x

Table 2: Cumulated survival rate (CSR, percentage) for MED-EL C40+, Pulsar, and Concerto devices implanted between 2003 and 2013. DR: device related failure, ACC: accident related failure, MED: medical reason for explant. Longest observation period (LOP): C40+ 10 years, Pulsar 9 years, Sonata 7 years, Mi1000 (Concerto) 2.5 years. Data extracted from [58].

Regarding the C40+ device, the manufacturer made a change in the production of the stimulator housing in 2003. Micro-leaks at the soldering point of the housing were frequently identified as the

ACCEPTED MANUSCRIPT

reason for the observation of moisture in devices that were analyzed after failure and explantation. Thick-film technology was replaced by thin-film laser soldering, which led to improved CSRs of the C40+ device after implementation.

In general, device-related (DR) failures in children were significantly more frequent than DR failures in adults. DR failure rate of the children is depending on device type. This is not unexpected. Head trauma is a common cause of device failure in children and the advanced impact resistance in newer implant generations (SYNCHRONY 2: impact resistance 2.5 Joule) has improved CSRs considerably.

3. Expert commentary

MED-EL's high-rate multichannel implant system has an exceptionally successful development history. Starting as a small start-up company with a handful of employees in the early 1990s, MED-EL has grown into an internationally renowned company with over 2,200 employees. The success is based on the numerous innovations that the company was able to bring to the market in the special field of CI systems. This is evidenced e.g. by the introduction of extra-long electrode carriers that allow stimulation of the apical areas of the cochlea, the introduction of the first high rate multichannel processor that can be worn behind the ear, the development of atraumatic electrode arrays, or the development of processing devices which combine electric and acoustic stimulation. MED-EL introduced the first 3 T MRI compatible implant magnet design to the market as well.

Not all innovations were initially able to meet the high demands of everyday life. For example, the cable connection between the processor unit and the transmitter coil had to be reworked several times in order to achieve the longest possible service life. The design of the processor also had to be adapted to prevent the penetration of body sweat. Likewise, measures were taken to improve the implant's hermeticity and to strengthen the implant's impact resistance. A particular strength of the company is the quick identification of the required improvement potential as well as the implementation of the necessary technical changes to improve their products.

In summary, the current MED EL cochlear implant system presents cutting-edge technology to provide a safe and effective device for hearing rehabilitation.

As far as further development is concerned, both the external and the internal part of the CI systems are constantly being technically improved. In the past, companies have made great efforts to reduce the size of the processor as well as of the stimulator. Future versions of the SONNET 2 processor could feature an even smaller profile to further increase wearing comfort. The development of direct Bluetooth streaming to the mobile phone is desirable without the need for an additional AUDIOLINK component, as has already been implemented by some competitors. Furthermore, a reduction in energy consumption and an increase in the capacity of the Me1510 processor's rechargeable batteries would be beneficial in order to achieve a minimum runtime of one day of use. The single unit Me1150 processor introduced 2017 could be upgraded within the next 5 years by integrating Bluetooth streaming and the dual-microphone system already integrated in the Me1510 processor. As with the Me1510 processor, the Me1150 processor should also be able to be controlled by a smartphone app.

Regarding surgery techniques, measurements for intra-operative monitoring of low frequency hearing function are expected to become part of clinical routine within the next five years [40]. Potentially, advanced methods will be used to assess the function of the auditory nerve, which will allow an immediate estimation of the expected outcome [60].

The application of robot-assisted surgery in CI provision is still the subject of current research (e.g. [61]). Studies are currently investigating whether access to the middle ear (posterior tympanotomy)

ACCEPTED MANUSCRIPT

by means of robot-guided drilling is technically feasible. Likewise, the robot-controlled opening of the inner ear as well as the automated insertion of the electrode array into the cochlea are topics of current research [62]. In view of the high demands on accuracy and safety of these robot-guided procedures, an implementation of these techniques within the next five years is very difficult to imagine.

Funding

This paper was not funded.

Declaration of interest

U Baumann, T Stöver and T Weissgerber were previously appointed as consultants for MED-EL. Several research cooperations have existed and currently exist between the company and the authors' institution. The authors have no other relevant affiliations or financial involvement with any organization or entity with a financial interest in or financial conflict with the subject matter or materials discussed in the manuscript apart from those disclosed.

Reviewer disclosures

One peer reviewer is paid by MED-EL as the Audiology Advisory Board member. Peer reviewers on this manuscript have no other relevant financial relationships or otherwise to disclose.

ACCEPTED MANUSCRIPT

References

[1] MED-EL I. Cochlear Implant Reliability [Internet] [cited 2020 Feb 9]. Available from: https://www.medel.com/en-gb/hearing-solutions/cochlear-implants/reliability.

[2] Quintern J, Jaeger RJ, Baumann U. Neural Prostheses. In: Neurological Disorders; 2003; p. 1199–1219.

[3] Wilson BS. The Remarkable Cochlear Implant and Possibilities for the Next Large Step Forward [Internet] [cited 2020 Feb 21]. Available from: https://acousticstoday.org/the-remarkable-cochlear-implant-and-possibilities-for-the-next-large-step-forward-blake-s-wilson/.

[4] MED-EL. About MED-EL: Our Timeline [Internet] [cited 2020 Feb 21]. Available from: https://www.medel.com/en-gb/about-medel/our-history.

[5] Bahmer A, Adel Y, Baumann U. Preventing Facial Nerve Stimulation by Triphasic Pulse Stimulation in Cochlear Implant Users: Intraoperative Recordings. Otol Neurotol. 2017;38:e438-e444.

[6] Bahmer A, Baumann U. The Underlying Mechanism of Preventing Facial Nerve Stimulation by Triphasic Pulse Stimulation in Cochlear Implant Users Assessed With Objective Measure. Otol Neurotol. 2016;37:1131–1137.

[7] Bahmer A, Baumann U. Effects of electrical pulse polarity shape on intra cochlear neural responses in humans: Triphasic pulses with cathodic second phase. Hear. Res. 2013;306:123–130.

[8] Leinung M, Loth A, Gröger M, et al. Cochlear implant magnet dislocation after MRI: Surgical management and outcome. Eur Arch Otorhinolaryngol. 2020;277:1297–1304.

[9] MED-EL Medical Electronics. SYNCHRONY Surgical Guideline [Internet]. Innsbruck [cited 2020 Feb 21]. Available from: https://s3.medel.com/documents/AW/AW32151_30_Surgical%20Guideline%20SYNCHRONY%20-%20EN%20English_Web.pdf.

[10] MED-EL Medical Electronics. Important Safety Information [Internet] [cited 2020 Feb 23]. Available from: https://www.medel.com/en-gb/important-safety-information.

[11] Muehloecker C, Nielsen SB, Jaeger A. Using the Insertion Test Device for choosing the best suited cochlear implant electrode variant. International Congress Series. 2004;1273:101–104.

[12] MED-EL Medical Electronics. Electrode Arrays: Designed for Atraumatic Implantation Providing Superior Hearing Performance [Internet] [cited 2020 Feb 21]. Available from: https://s3.medel.com/pdf/21617.pdf.

[13] Dazert S, Thomas JP, Buchner A, et al. Off the ear with no loss in speech understanding: comparing the RONDO and the OPUS 2 cochlear implant audio processors. Eur Arch Otorhinolaryngol. 2017;274:1391–1395.

**[14] Ilberg CA von, Baumann U, Kiefer J, et al. Electric-acoustic stimulation of the auditory system: A review of the first decade. Audiol. Neuro. Otol. 2011;16 Suppl 2:1–30.
Summary about the development of combined electric-acoustic stimulation

[15] Food and Drug Administration. Summary of Safety and Effectiveness Data: PMA P000025/S084: [Internet]. [place unknown]; 2016 [cited 2020 Feb 21]. Available from: https://www.accessdata.fda.gov/cdrh_docs/pdf/p000025s084b.pdf.

[16] MED-EL Medical Electronics. SONNET (Me1310) and SONNET EAS (Me1320) audio processors: User Manual [Internet]. Innsbruck. Available from: https://cochlearimplanthelp.files.wordpress.com/2019/05/med-el-sonnet-2-user-manual.pdf.

[17] MED-EL Medical Electronics. Accessories: WaterWear [Internet] [cited 2020 Feb 21]. Available from: https://www.medel.com/en-gb/hearing-solutions/accessories/waterwear.

ACCEPTED MANUSCRIPT

[18] Zierhofer CM, Inventor; MED-EL Elektromedizinische Geräte GmbH Innsbruck (AT), assignee. Multichannel Cochlear Implant with Neural Response Telemetry. United States Patent US 6,600,955 B1 2003 Jul 29.

**[19] Zeng F-G, Rebscher S, Harrison W, et al. Cochlear implants: system design, integration, and evaluation. IEEE Rev Biomed Eng. 2008;1:115–142.
Technical review about cochlear implant systems.

[20] Strahl SB, Dierker A, Spitzer P, et al. AutoART – A system for automatic determination of eCAP thresholds. In: Rahne T, editor Proceedings of the 21. Annual Meeting of the German Audiological Society; Halle a.d.S.; 2018.

[21] Wilson BS, Finley CC, Lawson, D. T., et al. Design and evaluation of a continuous interleaved sampling (CIS) processing strategy for multichannel cochlear implants. Journal of rehabilitation research and development. 1993;30:110–116.

[22] Zierhofer CM, Inventor; MED-EL Elektromedizinische Geräte GmbH Innsbruck (AT), assignee. Electrical nerve stimulation based on channel specific sampling sequences. United States Patent US 6,594,525 B1 2003 Jul 15.

[23] Müller J, Brill S, Hagen R, et al. Clinical trial results with the MED-EL fine structure processing coding strategy in experienced cochlear implant users. ORL J Otorhinolaryngol Relat Spec. 2012;74:185–198.

[24] Kleine Punte A, Bodt M de, Heyning PH van de. Long-term improvement of speech perception with the fine structure processing coding strategy in cochlear implants. ORL J Oto-Rhino-Lary. 2014;76:36–43.

[25] Riss D, Hamzavi J-S, Blineder M, et al. FS4, FS4-p, and FSP: a 4-month crossover study of 3 fine structure sound-coding strategies. Ear Hear. 2014;35:e272-81.

[26] Zierhofer CM, Schatzer R. Simultaneous intracochlear stimulation based on channel interaction compensation: analysis and first results. IEEE Trans Biomed Eng. 2008;55:1907–1916.

[27] Bader P, Kals M, Schatzer R, et al. Compensation for channel interaction in a simultaneous cochlear implant coding strategy. J. Acoust. Soc. Am. 2013;133:4124–4132.

[28] Müller J, Baumgartner W-D, Godey B, et al. The MED-EL SONATATI 100 cochlear implant: An evaluation of its safety in adults and children. Acta Otolaryngol. 2011;131:504–511.

[29] Kuzovkov V, Sugarova S, Yanov Y. The Mi1000 CONCERTO PIN cochlear implant: An evaluation of its safety and stability in adults and children. Acta Otolaryngol. 2016;136:236–240.

[30] Rader T, Baumann U, Stöver T, et al. Management of Cochlear Implant Electrode Migration. Otol Neurotol. 2016;37:e341-8.

[31] Leinung M, Helbig S, Adel Y, et al. The Effect of a Bone Groove Against Cochlear Implant Electrode Migration. Otol Neurotol. 2019;40:e511-e517.

[32] Gabrielpillai, Jennis, Burck I, Baumann U, et al. Incidence for Tip Foldover During Cochlear Implantation. Otol Neurotol. 2018;39:1115–1121.

[33] Gifford RH, Stecker GC. Binaural cue sensitivity in cochlear implant recipients with acoustic hearing preservation. Hear Res. 2020;390:preprint.

[34] Brockmeier, S. J., Peterreins M, Lorens A, et al. Music perception in electric acoustic stimulation users as assessed by the Mu.S.I.C. test. Adv Otorhinolaryngol. 2010;67:70–80.

*[35] Rader T, Fastl H, Baumann U. Speech perception with combined electric-acoustic stimulation and bilateral cochlear implants in a multisource noise field. Ear Hear. 2013;34:324–332.
Speech perception of MED-EL CI-users in complex listening tasks

[36] Helbig S, Adel Y, Leinung M, et al. Hearing Preservation Outcomes After Cochlear Implantation Depending on the Angle of Insertion: Indication for Electric or Electric-Acoustic Stimulation. Otol Neurotol. 2018;39:834–841.

[37] Skarzynski H, Heyning PH van de, Agrawal S, et al. Towards a consensus on a hearing preservation classification system. Acta Otolaryngol Suppl. 2013:3–13.

ACCEPTED MANUSCRIPT

[38] Snels C, IntHout J, Mylanus E, et al. Hearing Preservation in Cochlear Implant Surgery: A Meta-Analysis. Otol Neurotol. 2019;40:145–153.
[39] Canfarotta MW, Dillon MT, Buss E, et al. Validating a New Tablet-based Tool in the Determination of Cochlear Implant Angular Insertion Depth. Otol Neurotol. 2019;40:1006–1010.
[40] Giardina CK, Brown KD, Adunka OF, et al. Intracochlear Electrocochleography. Ear Hear. 2019;40:833–848.
[41] Fontenot TE, Giardina CK, Fitzpatrick DC. A Model-Based Approach for Separating the Cochlear Microphonic from the Auditory Nerve Neurophonic in the Ongoing Response Using Electrocochleography. Front Neurosci. 2017;11:592.
*[42] Helbig S, Adel Y, Rader T, et al. Long-term Hearing Preservation Outcomes After Cochlear Implantation for Electric-Acoustic Stimulation. Otol Neurotol. 2016;37:e353-9.
Long term EAS results
[43] Günther, Stefanie, Baumann U, Stöver T. Early Fitting in Cochlear Implantation: Benefits and Limits. Otol Neurotol. 2018;39:e250-e256.
[44] Pals C, Sarampalis A, Beynon A, et al. Effect of Spectral Channels on Speech Recognition, Comprehension, and Listening Effort in Cochlear-Implant Users. Trends Hear. 2020;24:1–15.
[45] Blamey P, Artieres F, Başkent D, et al. Factors affecting auditory performance of postlinguistically deaf adults using cochlear implants: an update with 2251 patients. Audiol. Neuro. Otol. 2013;18:36–47.
[46] Zeh R, Baumann U. Stationäre Rehabilitationsmaßnahmen bei erwachsenen CI-Trägern: HNO. HNO. 2015;63:557–576.
[47] Helms J, Müller J, Schön F, et al. Evaluation of Performance with the COMBI 40 Cochlear Implant in Adults: A Multicentric Clinical Study. ORL. 1997;59:23–35.
[48] Wimmer W, Weder S, Caversaccio M, et al. Speech Intelligibility in Noise With a Pinna Effect Imitating Cochlear Implant Processor. Otol Neurotol. 2016;37:19–23.
[49] Honeder C, Liepins R, Arnoldner C, et al. Fixed and adaptive beamforming improves speech perception in noise in cochlear implant recipients equipped with the MED-EL SONNET audio processor. PLoS ONE. 2018;13:e0190718.
[50] Hagen R, Radeloff A, Stark T, et al. Microphone directionality and wind noise reduction enhance speech perception in users of the MED-EL SONNET audio processor. Cochlear Implants Int. 2020;21:53–65.
[51] Weissgerber T, Stöver T, Baumann U. Speech perception in noise: Impact of directional microphones in users of combined electric-acoustic stimulation. PLoS ONE. 2019;14:e0213251.
[52] Limb, C. J., Roy, A. T. Technological, biological, and acoustical constraints to music perception in cochlear implant users. Hear Res. 2014;308:13–26.
[53] Zeng F-G, Tang Q, Lu T. Abnormal pitch perception produced by cochlear implant stimulation. PLoS ONE. 2014;9:e88662.
**[54] Rader T, Döge J, Adel Y, et al. Place dependent stimulation rates improve pitch perception in cochlear implantees with single-sided deafness. Hear. Res. 2016:94–103.
Pitch perception study in SSD subjects.
[55] Schatzer R, Vermeire K, Visser D, et al. Electric-acoustic pitch comparisons in single-sided-deaf cochlear implant users: frequency-place functions and rate pitch. Hear Res. 2014;309:26–35.
[56] Sharma SD, Cushing SL, Papsin BC, et al. Hearing and speech benefits of cochlear implantation in children: A review of the literature. Int J Ped ORL. 2020;133:109984.
[57] MED-EL. Zwei sind besser als Eines: 18 Jahre Hören mit zwei CIs [Internet] [cited 2020 May 3]. Available from: https://www.youtube.com/watch?v=4lr__rpQgIM.
*[58] Heyning PH van de, Atlas M, Baumgartner W-D, et al. The reliability of hearing implants: Report on the type and incidence of cochlear implant failures. Cochlear Implants Int. 2020:1–10. Latest report on reliability of MED-EL cochlear implants.

ACCEPTED MANUSCRIPT

[59] Wang JT, Wang AY, Psarros C, et al. Rates of revision and device failure in cochlear implant surgery: a 30-year experience. Laryngoscope. 2014;124:2393–2399.
[60] Lassaletta L, Polak M, Huesers J, et al. Usefulness of Electrical Auditory Brainstem Responses to Assess the Functionality of the Cochlear Nerve Using an Intracochlear Test Electrode. Otol Neurotol. 2017;38:e413–e420.
[61] Caversaccio M, Gavaghan K, Wimmer W, et al. Robotic cochlear implantation: Surgical procedure and first clinical experience. Acta Otolaryngol. 2017:1–11.
[62] Du X, Brett PN, Zhang Y, et al. A hand-guided robotic drill for cochleostomy on human cadavers. RSRR. 2018;Volume 5:13–18.

ACCEPTED MANUSCRIPT